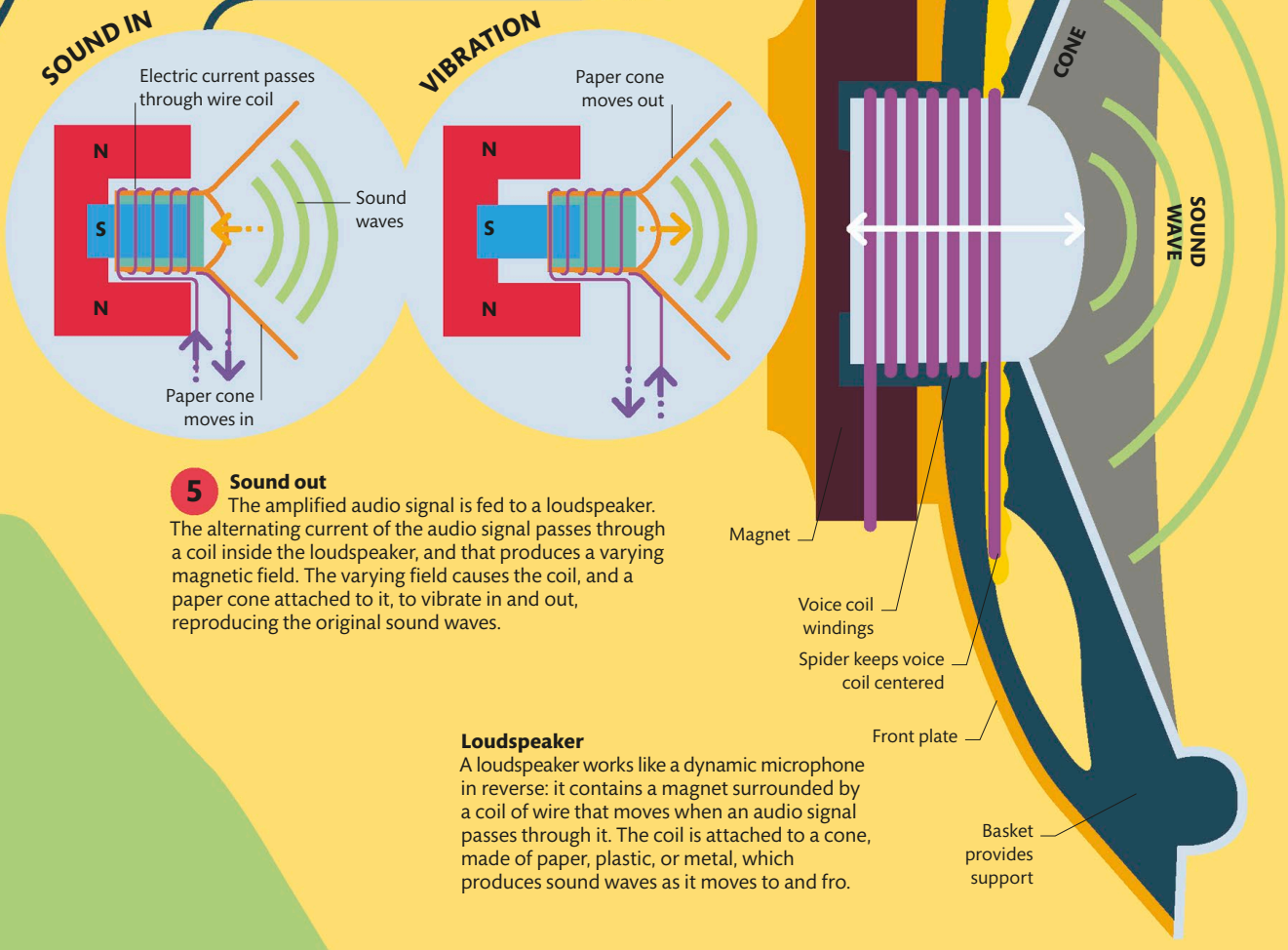


4 Signal amplification
The audio signal produced by a microphone is not powerful enough to produce sound in a loudspeaker. An electronic circuit called an amplifier boosts the signal.

Making sound

Loudspeakers use audio signals to reproduce sound. The audio signal might come straight from a microphone, or it might be output from the memory of a computer or smartphone where it has been stored. It might even arrive wirelessly, coded into radio waves. Wherever it comes from, it is too weak to produce a loud sound, and it must be strengthened (amplified) before reaching the loudspeaker.



5 Sound out
The amplified audio signal is fed to a loudspeaker. The alternating current of the audio signal passes through a coil inside the loudspeaker, and that produces a varying magnetic field. The varying field causes the coil, and a paper cone attached to it, to vibrate in and out, reproducing the original sound waves.

Loudspeaker
A loudspeaker works like a dynamic microphone in reverse: it contains a magnet surrounded by a coil of wire that moves when an audio signal passes through it. The coil is attached to a cone, made of paper, plastic, or metal, which produces sound waves as it moves to and fro.